

SECTION A: Conceptual Understanding

1. Why do we need autograd if we already know calculus and can compute derivatives manually?
 2. What problem does autograd solve in deep neural networks with many layers?
 3. What is a computation graph? Explain it using a simple example like:
 $y = (x^2 + 3x) * 2$
 4. In a computation graph, what do nodes represent and what do edges represent?
 5. Why is the computation graph called a Directed Acyclic Graph (DAG)?
 6. What happens to gradient flow if the graph contains a cycle?
 7. Explain in your own words how the chain rule is applied automatically in autograd.
-

SECTION B: Computation Graph and Chain Rule

8. Suppose:
 $x = 2$
 $y = x^2$
 $z = 3 * y$
 $L = z + 4$
Draw the computation graph and manually compute dL/dx .
9. If:
 $y = \text{sigmoid}(w * x + b)$
 $L = (y - t)^2$
Write the full chain rule expression for dL/dw before simplifying.
10. Why does Binary Cross Entropy with Sigmoid simplify to $(y_{\text{pred}} - y)$?
11. What would change in the gradient flow if we used ReLU instead of Sigmoid?